Recurrent architecture

Recurrent cell
RNN / GRU / LSTM / . . .

Capacity
Hidden-state size; number of layers

Directionality
Unidirectional / bidirectional

Decoder & regularization
Size; activations; dropout

Data representation

Plot-level inputs
Repeat in time / initialize hidden state

Categorical variables
One-hot encoding / embeddings / . . .

Time grid
Regular / irregular

Measurement methods
Micrometeorological only / multiple

One model
A combination
of choices

Supervision & output

Response
Cumulative / interval / flux

Supervision
Single response / joint responses

Response scaling
Raw / plot-wise (TAN) / global

Output construction
Direct cumulative / integrated flux

Training strategy

Loss
MSE / MAE / weighted / . . .

Optimization
Optimizer; learning rate; schedule

Training control
Batch size; early stopping

Data augmentation
None / augmentation strategy

Physical constraints

Constraint level
Unconstrained / constrained

Flux & cumulative emissions
Positive flux; monotonic cumulative

Available nitrogen
Upper bound based on applied TAN

Physical formulation
Output constraints / finite reservoir

Many possible combinations



. . .

Each column: one combination of choices.
Choices interact and influence predictive performance.
Exploring every combination is generally impractical.

Representative choices; schematic combinations. Some choices depend on others.